# Bayesian Analytics
# &
# Some Other Thoughts

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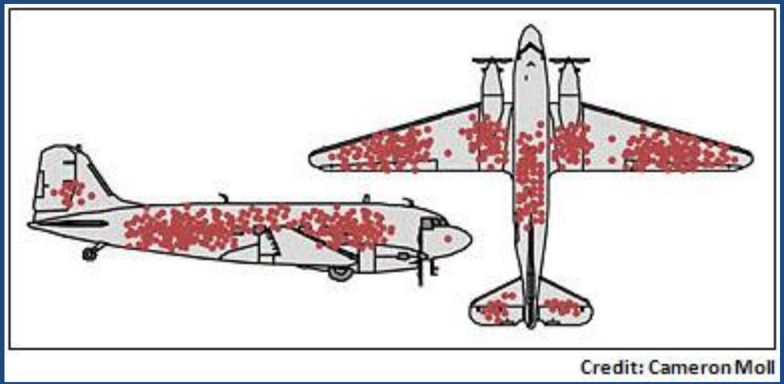
# Overview

- Abraham Wald’s Bomber Analysis
- Bayesian Statistical Inference
- Bayesian Model Averaging

# Abraham Wald's WWII Bomber Analysis

- Where to armor the bombers?
- Study of "hit analysis"
- Armor where bullets are NOT!




The Endeavor Blog (J. Cook)

# “Observation is Selection”

## Whitehead, 1925

- Data are not data until they are observed
- Observation is straightforward, but the selection process is not

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# The impact of missing trauma data on predicting massive transfusion

**Amber W. Trickey, PhD, CPH, Erin E. Fox, PhD, Deborah J. del Junco, PhD, Jing Ning, PhD, John B. Holcomb, MD, Karen J. Brasel, MD, MPH, Mitchell J. Cohen, MD, Martin A. Schreiber, MD, Eileen M. Bulger, MD, Herb A. Phelan, MD, MSCS, Louis H. Alarcon, MD, John G. Myers, MD, Peter Muskat, MD, Bryan A. Cotton, MD, MPH, Charles E. Wade, PhD, *and* Mohammad H. Rahbar, PhD, on behalf of the PROMMTT Study Group, *Houston, Texas***

**TABLE 3.** Summary of Missing Data for Injury Characteristics, Physiologic Measures, and Laboratory Values by MT Status and Survival at 6 Hours and 24 Hours After ED Admission

| | Missing by MT | | | Missing by Survival | | | | Missing by Study Site | |
|---|---|---|---|---|---|---|---|---|---|
| | MT Missing, n (%) | Non-MT Missing, n (%) | $p$* | Survived >24 h Missing, n (%) | Survived 6–24 h Missing, n (%) | Death <6 h Missing, n (%) | $p$* | Missing Range Among 10 Sites, % | $p$* |
| Total, n | 295 | 939 | — | 1,097 | 46 | 102 | — | — | — |
| GCS total | 39 (13) | 70 (7) | 0.003 | 97 (9) | 2 (4) | 11 (11) | 0.45 | 2–21 | <0.001 |
| SBP | 21 (7) | 10 (1) | <0.001 | 16 (1) | 3 (7) | 13 (13) | <0.001 | 0–7 | <0.001 |
| DBP | 68 (23) | 148 (16) | 0.004 | 176 (16) | 9 (20) | 32 (31) | <0.001 | 0–75 | <0.001 |
| Heart rate | 13 (4) | 13 (1) | 0.003 | 19 (2) | 0 (0) | 8 (8) | 0.002 | 0–7 | <0.001 |
| Respiratory rate | 149 (50) | 409 (43) | 0.04 | 455 (41) | 35 (76) | 69 (68) | <0.001 | 10–83 | <0.001 |
| Hematocrit | 20 (7) | 30 (3) | 0.01 | 31 (3) | 3 (7) | 13 (13) | <0.001 | 0–11 | <0.001 |
| Hemoglobin | 19 (6) | 27 (3) | 0.01 | 31 (3) | 3 (7) | 13 (13) | <0.001 | 0–9 | <0.001 |
| Base deficit | 62 (21) | 223 (24) | 0.34 | 243 (22) | 9 (20) | 33 (32) | 0.06 | 6–90 | <0.001 |
| pH | 59 (20) | 210 (22) | 0.38 | 232 (21) | 8 (17) | 30 (29) | 0.12 | 0–90 | <0.001 |

*$p$ values were calculated using Pearson's $\chi^2$ tests or Fisher's exact tests when $\chi^2$ assumptions were not met.
DBP, diastolic blood pressure; GCS, Glasgow Coma Scale; MT, massive transfusion; SBP, systolic blood pressure.

TABLE 6. Comparison of MT Predictive Model Capability Using Complete Case Analysis and Imputation

| Algorithm | Analysis | Dataset | n (%) | Sensitivity, % | Specificity, % | Correctly Classified, % | AUC |
|---|---|---|---|---|---|---|---|
| McLaughlin et al.[21] | | | | | | | |
| | Complete case | PROMMTT | 936 (75) | 65 | 58 | 60 | 0.62 |
| | Multiple imputation | PROMMTT | 1,198 (96) | 61 | 62 | 62 | 0.62 |
| | Upper bound | PROMMTT | 1,245 (100) | 66 | 63 | 63 | 0.64 |
| | Lower bound | PROMMTT | 1,245 (100) | 57 | 60 | 59 | 0.59 |
| | Original report | McLaughlin et al.[21] | 396 (NR)* | 59 | 77 | 70 | 0.75 |
| Cancio et al.[23] | | | | | | | |
| | Complete case | PROMMTT | 513 (41)** | 77 | 32 | 41 | 0.54 |
| | Multiple imputation | PROMMTT | 1,120 (90) | 77 | 30 | 40 | 0.54 |
| | Upper bound | PROMMTT | 1,245 (100) | 82 | 32 | 46 | 0.57 |
| | Lower bound | PROMMTT | 1,245 (100) | 76 | 27 | 36 | 0.51 |
| | Original report | Cancio et al.[23] | 536 (77) | NR | NR | 62 | 0.64 |
| Larson et al.[24] | | | | | | | |
| | Complete case | PROMMTT | 1,095 (88)† | 82 | 43 | 53 | 0.63 |
| | Multiple imputation | PROMMTT | 1,219 (98) | 80 | 45 | 53 | 0.63 |
| | Upper bound | PROMMTT | 1,245 (100) | 84 | 50 | 58 | 0.67 |
| | Lower bound | PROMMTT | 1,245 (100) | 73 | 38 | 46 | 0.55 |
| | Original report | Larson et al.[24] | 1,124 (53) | 69 | 65 | 66 | 0.67 |

*Predictive capability was calculated using a validation data set, separate from the data set used for model development. Incomplete cases were excluded from the validation set. The number of incomplete cases is not provided.

**One subject had SBP = 0 and HR = 0; SI could not be calculated.

†Complete cases must contain at least two positively coded variables or three negatively coded variables for evaluation, as required by the Larson et al.[24] algorithm.

AUC, area under the receiver operating characteristic curve; MT, massive transfusion; NR, not reported; SBP, systolic blood pressure; SI, shock index.

# Incidence, Predictors, and Outcome of Difficult Mask Ventilation Combined with Difficult Laryngoscopy

*A Report from the Multicenter Perioperative Outcomes Group*

**Table 1.** Univariate Patient Comparison for the Entire Study Population

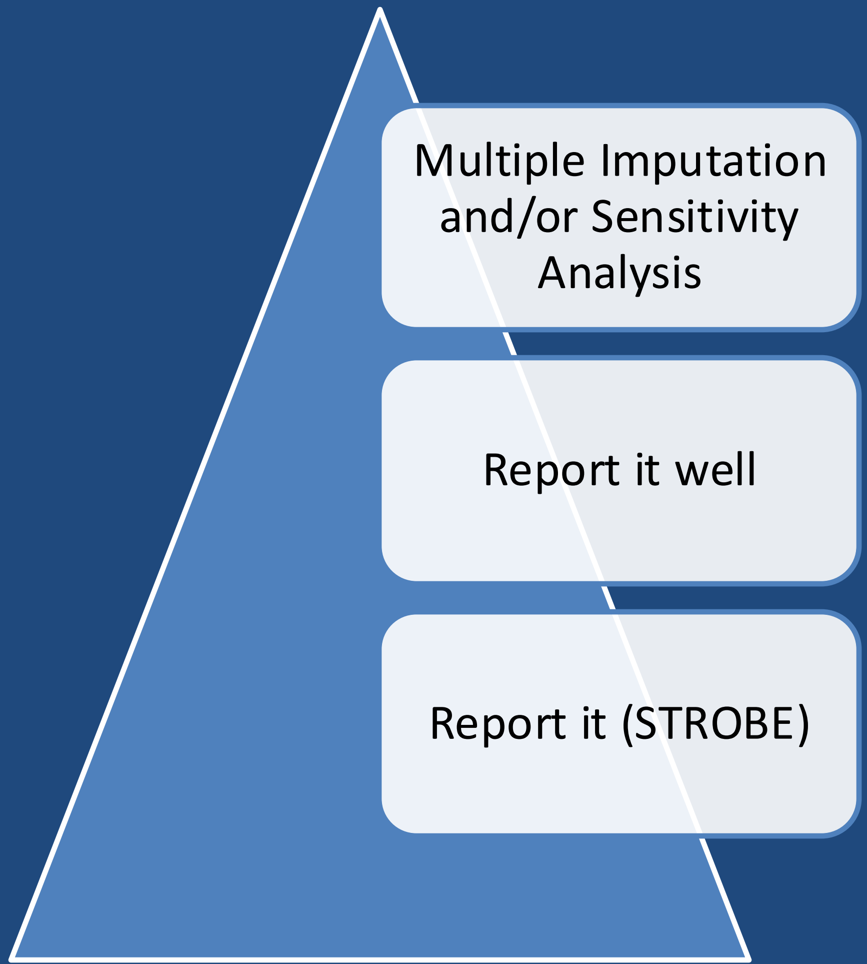
| | DMV Combined with DL No (N = 175,981) | DMV Combined with DL Yes (N = 698) | *P* Value | Odds Ratio (95% CI) | Percent Complete Date (%)† |
|---|---|---|---|---|---|
| Age ≥46 | 118,227 (67%) | 596 (85%) | <0.001 | 2.9 (2.3–3.5) | 100 |
| Body mass index (kg/k$^2$) ≥30 | 49,130 (35%) | 373 (69%) | <0.001 | 4.0 (3.3–4.8) | 82 |
| Male sex | 82,700 (47%) | 534 (77%) | <0.001 | 3.7 (3.1–4.4) | 100 |
| Limited thyromental distance | 10,407 (6.8%) | 101 (17%) | <0.001 | 2.8 (2.2–3.4) | 94 |
| Mallampati III or IV | 26,529 (17%) | 308 (49%) | <0.001 | 4.7 (4.0–5.5) | 93 |
| Presence of beard | 14,751 (13%) | 146 (30%) | <0.001 | 3.0 (2.5–3.6) | 80 |
| Sleep apnea | 25,393 (17%) | 268 (45%) | <0.001 | 4.0 (3.4–4.7) | 90 |
| Presence of teeth | 127,682 (83%) | 545 (91%) | <0.001 | 2.0 (1.5–2.6) | 96 |
| Limited or severely limited jaw protrusion | 10,811 (8.3%) | 98 (18%) | <0.001 | 2.5 (2.0–3.1) | 88 |
| Thick neck | 22,128 (13%) | 265 (38%) | <0.001 | 4.7 (4.0–5.5) | 100 |
| Neck mass or radiation | 1,542 (1.0%) | 18 (3.0%) | <0.001 | 3.1 (1.9–4.9) | 96 |
| Limited mouth opening | 5,349 (3.5%) | 49 (8.1%) | <0.001 | 2.4 (1.8–3.3) | 96 |
| Snoring | 54,725 (31%) | 367 (53%) | <0.001 | 2.5 (2.1–2.9) | 100 |
| Center 1* | 48% | 46% | 0.131 | 0.9 (0.8–1.0) | 100 |
| Center 2* | 20% | 14% | <0.001 | 0.6 (0.5–0.8) | 100 |
| Center 3* | 15% | 24% | <0.001 | 1.9 (1.6–2.2) | 100 |
| Center 4* | 17% | 16% | 0.686 | 1.0 (0.8–1.2) | 100 |

* To ensure the blinding of specific center and outcomes, absolute sample sizes for each centers are not included. The percent data reflect the distribution of patients by center (*i.e.*, center 2 represented 20% of the patients without the primary outcome). † Percent data complete is based on the derivation data for centers 1, 2, and 3. Center 4 did not include beard and jaw protrusion as a standard airway examination element.

DL = difficult laryngoscopy; DMV = difficult mask ventilation.

# What to do about missing data?

- Report it (STROBE)
  - “Explain how missing data were addressed”
- Report it well
  - Predictors of missing-ness
- Imputation
  - Likelihood-based approaches
  - Weighted estimation
  - Multiple imputation
- Sensitivity analyses
  - What if?
  - Best case, worst case



# Bayesian Statistical Inference

- We are in the midst of a revolution (could you tell?)
- Software improvements developing rapidly, allowing access to modeling previously thought untenable
- A complete introduction to Bayesian statistical inference can be found in many places, I'd like to show you some unique aspects of the approach

# Quick Comparison

**Frequentist**

- Conditions on the null hypothesis
  – P (data | hypothesis)
- Data as random and parameters as fixed
- Probability as long run frequency

**Bayesian**

- Conditions on the data
  – P (hypothesis | data)
- Data as fixed, parameters as random
- Probability as subjective combination of previous beliefs and current data

# MPOG study

- The MPOG completes a study and finds that a change in practice improves survival

- N = 180,000

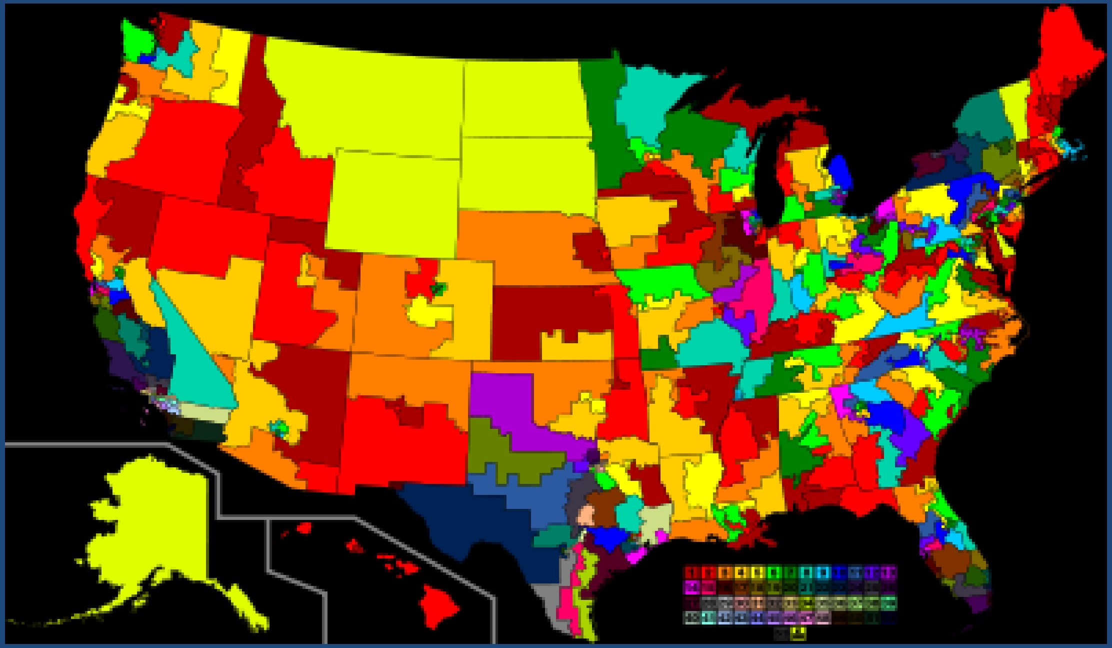
- 35% <u>improved</u> survival from standard practice

# Two Reviewers:

- **<u>A Skeptic</u>**
- “I seriously doubt the validity of this practice. I also hate the researchers.”

- **<u>A Supporter</u>**
- “This practice is going to improve everything. I’m proud of these special researchers”

# Nate Silver 2012 Elections



Presidential Election

32/33 Senatorial Elections

~427/435 Congressional Elections

Nate Silver combines the predictions of many different polls by weighting their recency and accuracy

This is a form of Bayesian Model Averaging

# Prediction Models

- Theory Based
  - Select predictors based on a priori candidacy
- Univariate Screening
  - Examine all predictors for $p < 0.25$, include candidates in final model
- Data driven (a.k.a., "step-wise")
  - Optimize a selection criteria (e.g., p- value)
- Model Selection Criteria
  - Akaike Information Criteria, Bayesian Information Criteria

# Prediction Models

- Selecting one model from a pool of potential models and using it exclusively has risk
- There is uncertainty in model selection, but this is not usually reflected in the estimates
  - Much of the available information was used to specify which variables to include in the model
  - We acknowledge parameter uncertainty, but not model uncertainty
- A “Quiet Scandal” in statistics (Breiman, 1992)

# Bayesian Model Averaging: Rationale

- What if we could combine the predictions from several (many) different models to improve prediction?

- Any such effort would need to consider:
  - model uncertainty
  - Large dimensions (especially for MPOG)

# Bayesian Model Averaging: Example